Statistical analysis of experimental data
Least-squares method

Aleksander Filip Żarnecki

FACULTY OF PHYSICS
UNIVERSITY OF WARSAW

Lecture 07

November 19, 2026

Least-squares method

1. χ^2 distribution
2. Hypothesis Testing
3. Linear Regression
4. Homework

Maximum Likelihood Method (lecture 5)

For N independent measurements of $\mathbf{x}$, the product:

$$L = \prod_{j=1}^{N} f(\mathbf{x}^{(j)}; \boldsymbol{\lambda})$$

where PDF of $\mathbf{x}$ depends on the parameter set $\boldsymbol{\lambda}$, is called a likelihood function.

The most commonly used approach to parameter estimation is the maximum likelihood approach: as the best estimate of the parameter $\boldsymbol{\lambda}$ we choose the parameter values for which the likelihood function has a (global) maximum. Frequently used is also log-likelihood function:

$$\ell = \ln L = \sum_{j=1}^{N} \ln f(\mathbf{x}^{(j)}; \boldsymbol{\lambda})$$

we can look for maximum value of ℓ or minimum of $-2\,\ell = -2 \ln L$

Multiple parameter estimate

Likelihood function (and log-likelihood) can depend on multiple parameters:

$$\boldsymbol{\lambda} = (\lambda_1 \dots \lambda_p) \qquad L = \prod_{j=1}^{N} f(\mathbf{x}^{(j)}; \boldsymbol{\lambda}) \qquad \ell = \sum_{j=1}^{N} \ln f(\mathbf{x}^{(j)}; \boldsymbol{\lambda})$$

Best estimate of $\boldsymbol{\lambda}$, for given set of experimental results $\mathbf{x}^{(j)}$, corresponds to maximum of the likelihood function, which can be found by solving a system of equations:

$$\left.\frac{\partial \ell}{\partial \lambda_i}\right|_{i=1\dots p} = 0$$

The Likelihood Principle

G. Bohm and G. Zech

Given a p.d.f. $f(\mathbf{x}; \boldsymbol{\lambda})$ containing an unknown parameters of interest $\boldsymbol{\lambda}$ and observations $\mathbf{x}^{(j)}$, all information relevant for the estimation of the parameters $\boldsymbol{\lambda}$ is contained in the likelihood function $L(\boldsymbol{\lambda}; \mathbf{x}) = \prod f(\mathbf{x}^{(j)}; \boldsymbol{\lambda})$.

Parameter covariance matrix

For the considered case of multivariate normal distribution, best parameter estimates $\hat{\boldsymbol{\lambda}}$ are given by the measured variable values $\mathbf{x}$.

Unlike parameters $\boldsymbol{\lambda}$, parameter estimates $\hat{\boldsymbol{\lambda}}$ are random variables (functions of $\mathbf{x}$) and so we can consider covariance matrix for $\hat{\boldsymbol{\lambda}}$:

$$\mathbb{C}_{\mathbf{x}} = \mathbb{C}_{\hat{\boldsymbol{\lambda}}} = \left(-\frac{\partial^2 \ell}{\partial \lambda_i \, \partial \lambda_j} \right)^{-1}$$

Knowing the likelihood function, we can not only estimate parameter values, but also extract uncertainties and correlations of these estimates!

Recipe for a parameter uncertainty

G. Bohm and G. Zech

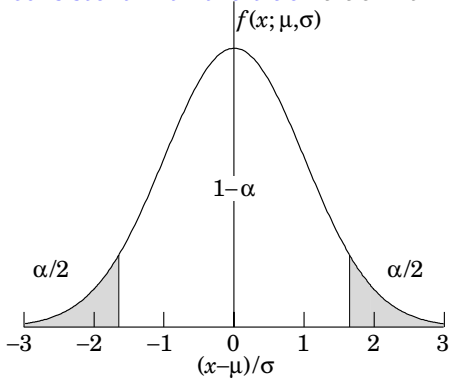
Standard error intervals of the extracted parameter are defined by the decrease of the log-likelihood function by 0.5 for one, by 2 for two and by 4.5 for three standard deviations.

Normal distribution

Meaning of σ is well defined for Gaussian distribution.

Probability for the experimental result to be consistent with the true value within $\pm N\,\sigma$:

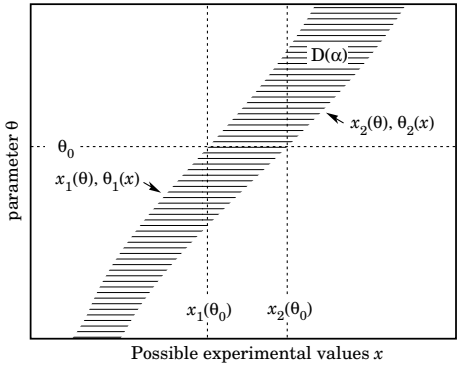
		$1-\alpha$	
$\pm 1\,\sigma$	$\Rightarrow$	68.27	%
$\pm 2\,\sigma$	$\Rightarrow$	95.45	%
$\pm 3\,\sigma$	$\Rightarrow$	99.73	%
$\pm 4\,\sigma$	$\Rightarrow$	99.9937	%
$\pm 5\,\sigma$	$\Rightarrow$	99.999943	%



There is a non-zero chance for deviation grater than 5σ, but it is extremely small

Frequentist confidence intervals

General procedure

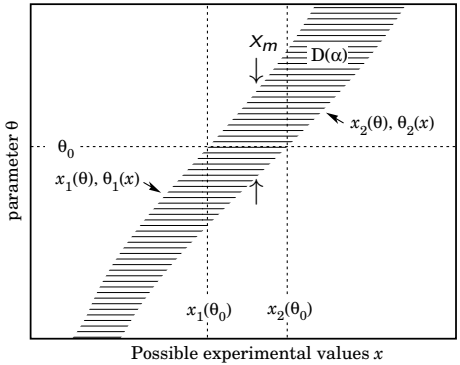


- calculate limits of probability intervals for x, $x_1(\theta)$ and $x_2(\theta)$, for different values of θ
- calculated intervals define the "accepted region" in (θ, x)

R.L. Workman et al. (Particle Data Group), Prog. Theor. Exp. Phys. 2022, 083C01 (2022), PDG web page

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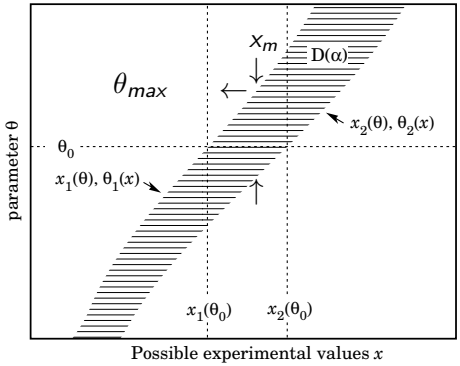


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$\Rightarrow$ limit on θ for given x_m, $\theta_1(x_m)$, corresponds to limit on x for given θ: $x_m = x_1(\theta_1)$.

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Maximum Likelihood Method

Let us consider N independent measurements of variable Y. Assuming **measurement fluctuations are described by Gaussian pdf**, the likelihood function is:

$$L = \prod_{i=1}^{N} G(y_i; \mu_i, \sigma_i) = \prod_{i=1}^{N} \frac{1}{\sigma_i \sqrt{2\pi}} \exp\left(-\frac{1}{2}\frac{(y_i - \mu_i)^2}{\sigma_i^2}\right)$$

Log-likelihood: assuming σ_i are known

$$\ell = -\frac{1}{2}\sum \frac{(y_i - \mu_i)^2}{\sigma_i^2} + \text{const}$$

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We can also define

$$\chi^2 = -2\,\ell = -2 \ln L = \sum_{i=1}^{N} \frac{(y_i - \mu_i)^2}{\sigma_i^2}$$

Maximum of (log-)likelihood function corresponds to minimum of χ^2 (for Gaussian pdf **!!!**)

χ^2 distribution

Problem

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It is a function of random variables, results of the measurement
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Can we predict what its probability distribution is?

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We will address this problem in two different approaches:

- simple one, based on intuitive case of $N = 2$, extrapolating to other N
- more formal one, based on momentum generating functions

N=2

Let us introduce "shift" variables:

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and then change variables to polar coordinates

see lecture 04

$$f(r_z, \phi_z) = \frac{1}{2\pi}\, r_z \exp\left(-\frac{1}{2} r_z^2\right)$$

N=2

Integrating over ϕ_z and changing variable to r_z^2

$$f(r_z^2) = \frac{1}{2} \exp\left(-\frac{1}{2} r_z^2\right)$$

Distribution is exponential, corresponds to decay time distribution for $\tau = 2$...

N=2

Integrating over ϕ_z and changing variable to $r_z^2 = \chi^2$

$$f(\chi^2) = \frac{1}{2} \exp\left(-\frac{1}{2}\chi^2\right)$$

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Extrapolation to even N case

Sum of $n = N/2$ numbers from exponential distribution, is distributed according to Gamma distribution with $k = n = N/2$, $\lambda = 1/\tau = 1/2$ lecture 03

$$f(\chi^2) = \frac{1}{\Gamma(k)} (\chi^2)^{k-1} \lambda^k e^{-\lambda\chi^2}$$

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$$f(\chi^2) = \frac{1}{\Gamma(\frac{N}{2})} \left(\frac{1}{2}\right)^{\frac{N}{2}} (\chi^2)^{\frac{N}{2}-1} e^{-\chi^2/2}$$

The formula (as one can expect) works also for odd N...

Moment generating function

Bonamente

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&= \frac{1}{\sqrt{1-2t}}
\end{aligned}
$$

Arbitrary N

Bonamente

Considered random variables z_i are independent, and $\chi^2 = \sum z_i^2$. Moment generating function for χ^2 distribution is thus given by:

$$M_N(t) = \prod_{i=1}^{N} M_1(t) = \left(\frac{1}{1-2t}\right)^{N/2}$$

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We can compare it with the moment generating functions for Gamma pdf

$$M_G(t) = \mathbb{E}(e^{tx}) = \int_0^{+\infty} dx \, \frac{1}{\Gamma(k)} x^{k-1} \, \lambda^k \, e^{-\lambda x} \, e^{tx}$$

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χ^2 distribution

We conclude that distribution of χ^2 is described by Gamma pdf with:

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Properties of the χ^2 distribution (see lecture 03)

$$\langle \chi^2 \rangle = \frac{k}{\lambda} = N$$

$$\mathbb{V}(\chi^2) = \frac{k}{\lambda^2} = 2N$$

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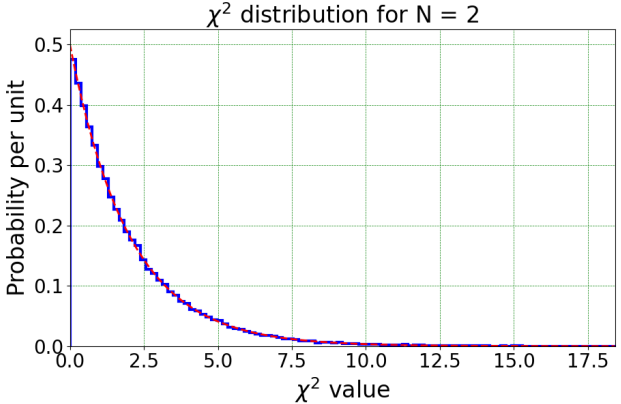
$$\sqrt{\mathbb{V}(\chi^2)} = \sigma_{\chi^2} = \sqrt{2N}$$

For small N, value of χ^2 is a subject to large fluctuations...

χ^2 distribution

07_chi2.ipynb

Results of the Monte Carlo sample generation (compared with predictions)

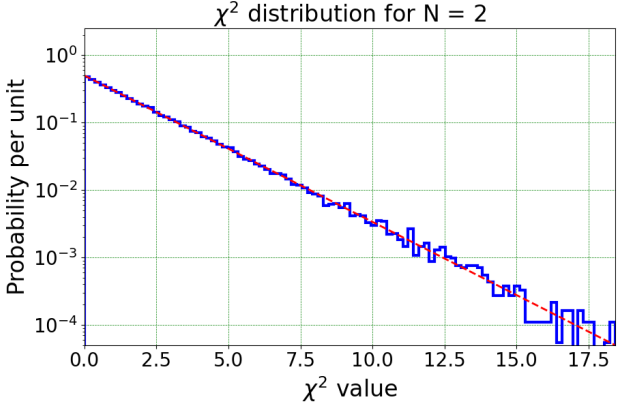


Exponential distribution

χ^2 distribution

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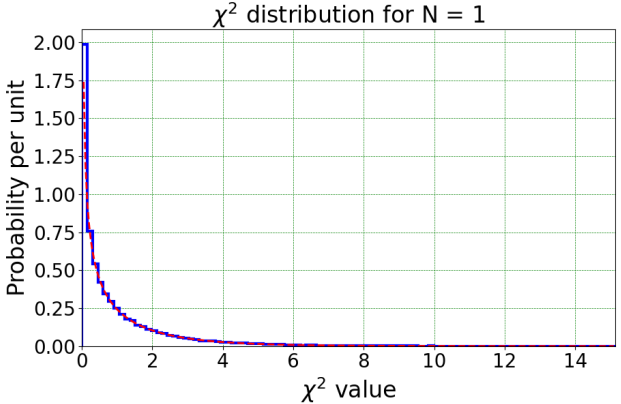


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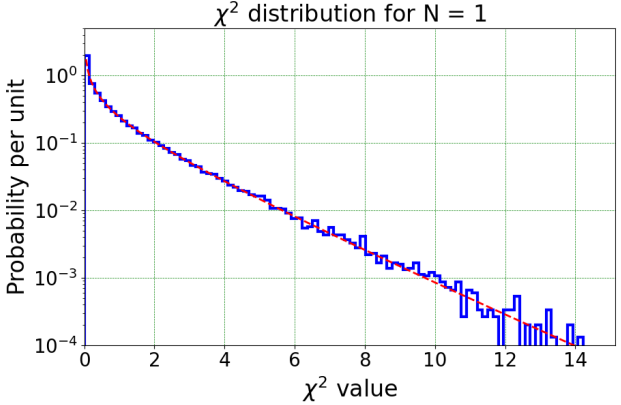
Sharply peaked at zero, but with long tail

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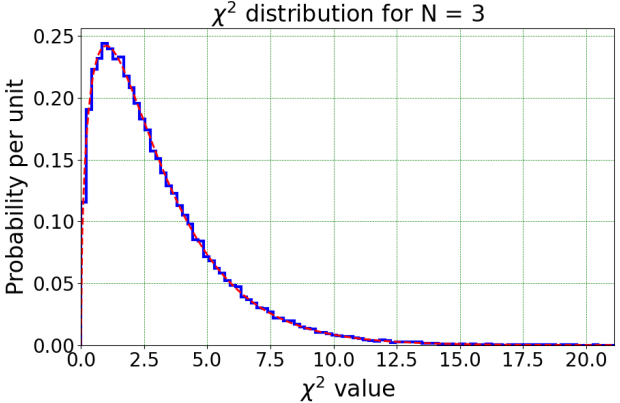


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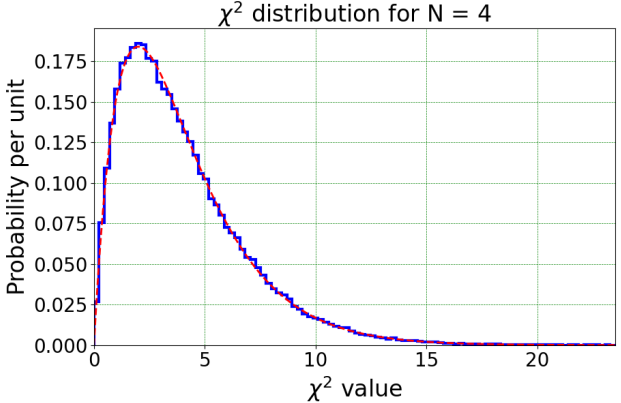


Very asymmetric, most events below average value...

χ^2 distribution

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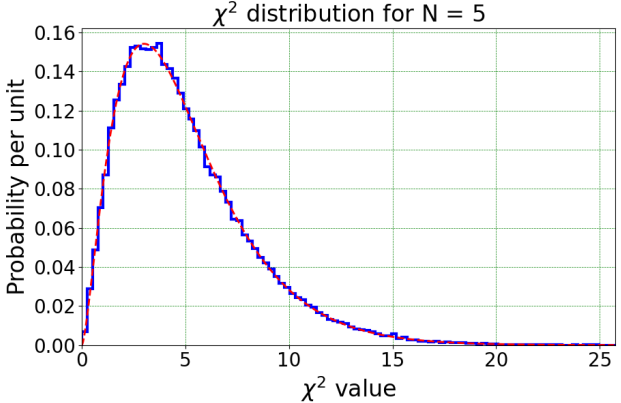


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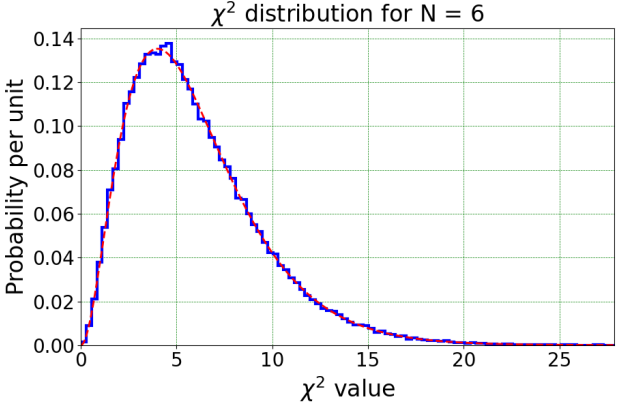


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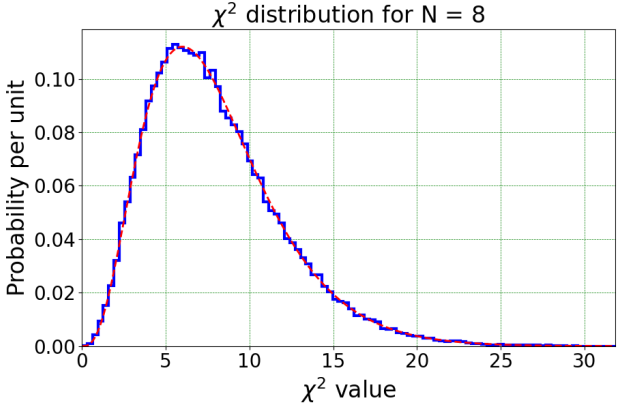


It is interesting to note that maximum position, $\chi^2_{max} = N - 2$ (!!!)

χ^2 distribution

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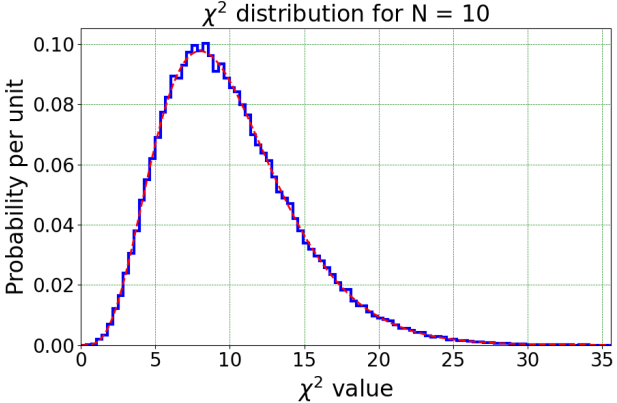


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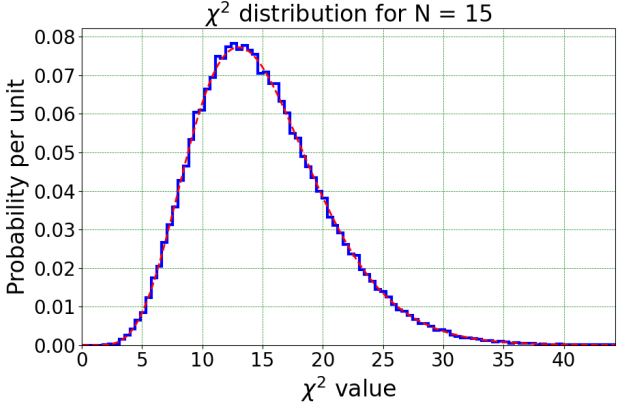


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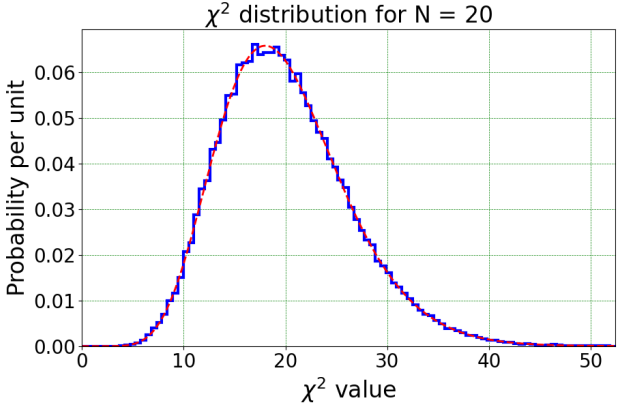
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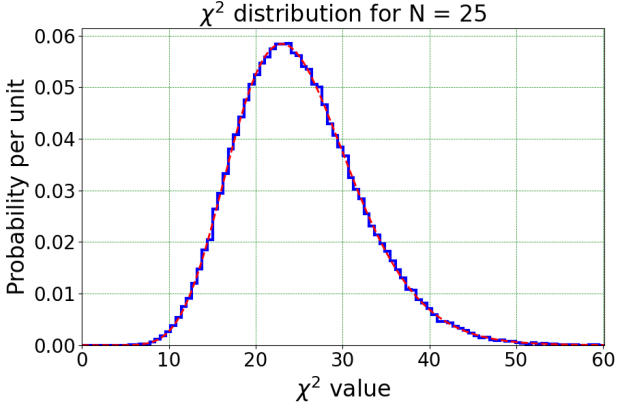
Results of the Monte Carlo sample generation (compared with predictions)



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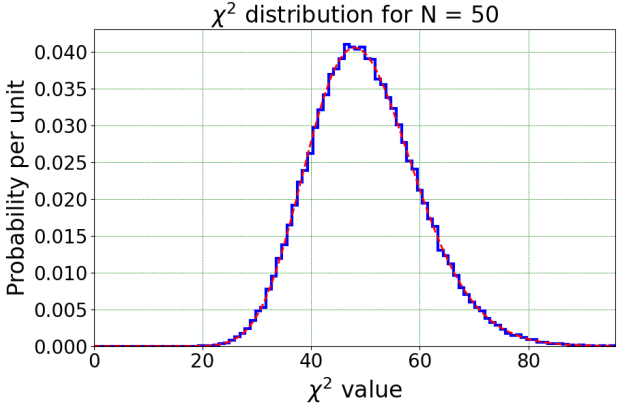
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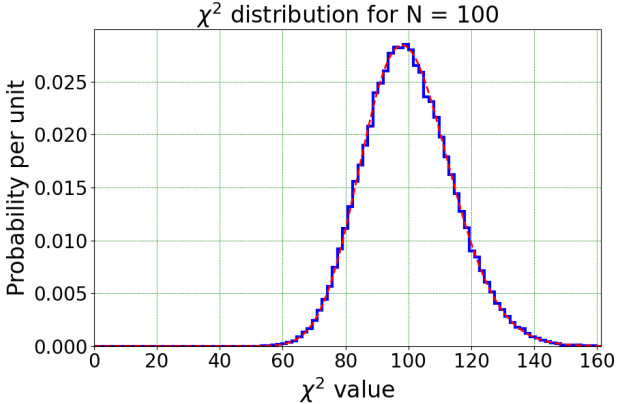
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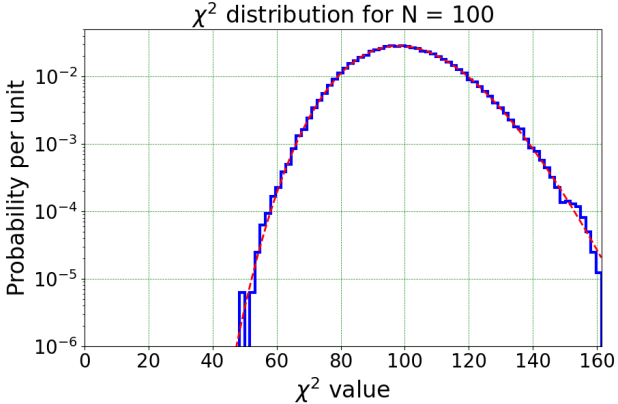
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Results of the Monte Carlo sample generation (compared with predictions)



Almost gaussian, but asymmetry in tails remains even for large N...

Reduced χ^2

When discussing consistency of large data samples it is often convenient to use value of "reduced χ^2":

$$\chi^2_{red} = \frac{\chi^2}{N}$$

Distribution of χ^2_{red} is again described by the Gamma pdf with

$$k = \frac{N}{2} \quad \text{and} \quad \lambda = \frac{N}{2}$$

Reduced χ^2

When discussing consistency of large data samples it is often convenient to use value of "reduced χ^2":

$$\chi^2_{red} = \frac{\chi^2}{N}$$

Distribution of χ^2_{red} is again described by the Gamma pdf with

$$k = \frac{N}{2} \quad \text{and} \quad \lambda = \frac{N}{2}$$

Properties of the distribution:

$$\langle \chi^2_{red} \rangle = 1 \qquad \mathbb{V}(\chi^2_{red}) = \sigma^2_{\chi^2_{red}} = \frac{2}{N} \qquad \chi^2_{red}\Big|_{\text{p=max}} = 1 - \frac{2}{N} \quad (N > 2)$$

Number of degrees of freedom

So far, we have only considered an ideal case, where both the expected values $\boldsymbol{\mu}$ and measurement uncertainties $\boldsymbol{\sigma}$ are known.

However, it is quite a common situation, when the expected value is extracted from the data:

$$\tilde{\chi}^2 = \sum_{i=1}^{N} \frac{(y_i - \bar{y})^2}{\sigma^2}$$

where we assume uniform uncertainties for simplicity.

What is the expected distribution for $\tilde{\chi}^2$?

Mean value corresponds to maximum likelihood $\Rightarrow$ $\tilde{\chi}^2 \leq \chi^2$

Number of degrees of freedom

We already know (lecture 04) that unbiased variance estimate for N measurements is

$$s^2 = \frac{1}{N-1}\sum_i (y_i - \bar{y})^2$$

so one can conclude:

$$\langle \tilde{\chi}^2 \rangle = N - 1$$

but this does not give us full information about the distribution...

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Simple variable transformation can be used: (Brandt)

$$x_1 = \frac{1}{\sqrt{2}}(y_1 - y_2)$$

$$x_2 = \frac{1}{\sqrt{2 \cdot 3}}(y_1 + y_2 - y_3)$$

$$\dots$$

Number of degrees of freedom

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Simple variable transformation can be used: (Brandt)

$$x_k = \frac{1}{\sqrt{k(k+1)}}(y_1 + \ldots + y_k - y_{k+1}) \quad k = 1 \ldots N-1$$

$$x_N = \sqrt{N} \cdot \bar{y}$$

One can verify that this is an orthogonal transformation...

Number of degrees of freedom

If y_i are independent random variables with Gaussian pdf, so are x_i. Also:

$$\sum_{i=1}^{N} x_i^2 = \sum_{i=1}^{N} y_i^2$$

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$$\sum_{i=1}^{N} x_i^2 = \sum_{i=1}^{N} y_i^2$$

We can now rewrite the formula for $\tilde{\chi}^2$ in the new basis:

$$\begin{aligned} \sigma^2 \cdot \tilde{\chi}^2 &= \sum_{i=1}^{N} (y_i - \bar{y})^2 = \sum_{i=1}^{N} y_i^2 - 2\bar{y} \sum_{i=1}^{N} y_i + N\bar{y}^2 \\ &= \sum_{i=1}^{N} y_i^2 - N\bar{y}^2 \end{aligned}$$

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$\Rightarrow$ distribution of $\tilde{\chi}^2$ corresponds to that of χ^2 for $N_{df} = N - 1$ variables...

Least-squares method

1. χ^2 distribution
2. Hypothesis Testing
3. Linear Regression
4. Homework

Data consistency test

Value of χ^2 (or χ^2_{red}) can be used to verify the consistency of the given data set $\mathbf{y}$ (with uncertainties $\boldsymbol{\sigma}$) with the model predictions given by $\boldsymbol{\mu}$

We can try to test the theoretical model, verify our estimates of measurement uncertainties, or check the consistency of the experimental procedure...

If the model does not describe the data, higher χ^2 values are expected.

How to quantify the level of agreement?

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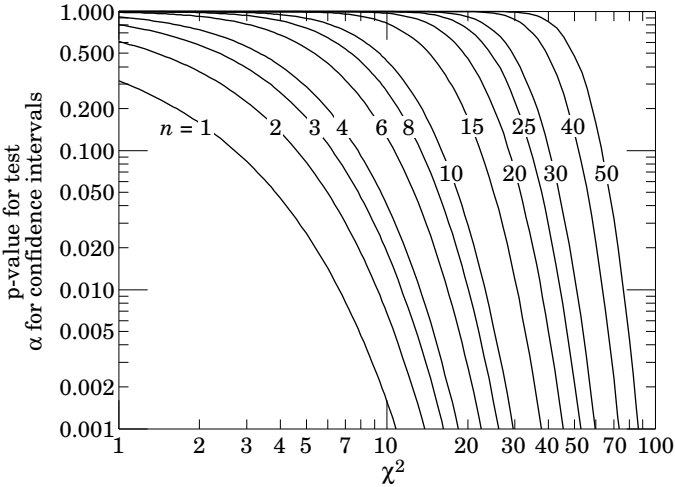
We can calculate the probability of obtaining given value of χ^2 or lower:

$$P(\chi^2) = \int_0^{\chi^2} d\chi^{2\prime}\, f(\chi^{2\prime})$$

given by the cumulative probability distribution. $1 - P(\chi^2)$ is sometimes referred to as p-value

Data consistency test

Plot of p-values as a function of χ^2 for different N_{df}



(PDG)

Critical χ^2

The other approach is to define, for given probability CL (confidence level) the critical value of χ^2, corresponding the the frequentist upper limit:

$$P(\chi^2_{crit}) = \int_0^{\chi^2_{crit}} d\chi^2 \, f(\chi^2) = CL \qquad \int_{\chi^2_{crit}}^{+\infty} d\chi^2 \, f(\chi^2) = 1 - CL = \alpha$$

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Very low P values, $P(\chi^2) \ll 1$, are also not expected (not likely)!
If $\chi^2 \ll N$ (except for very small N), this usually indicates a problem:

- overestimated uncertainties of measurements (or correlations not properly included)
- hidden correlations between measurements (which we treat as independent variables)

Critical χ^2

Table of critical χ^2 values (Brand)

$$P = \int_0^{\chi^2_P} f(\chi^2; f)\,\mathrm{d}\chi^2$$

	P				
f	0.900	0.950	0.990	0.995	0.999
1	2.706	3.841	6.635	7.879	10.828
2	4.605	5.991	9.210	10.597	13.816
3	6.251	7.815	11.345	12.838	16.266
4	7.779	9.488	13.277	14.860	18.467
5	9.236	11.070	15.086	16.750	20.515
6	10.645	12.592	16.812	18.548	22.458
7	12.017	14.067	18.475	20.278	24.322
8	13.362	15.507	20.090	21.955	26.124
9	14.684	16.919	21.666	23.589	27.877
10	15.987	18.307	23.209	25.188	29.588

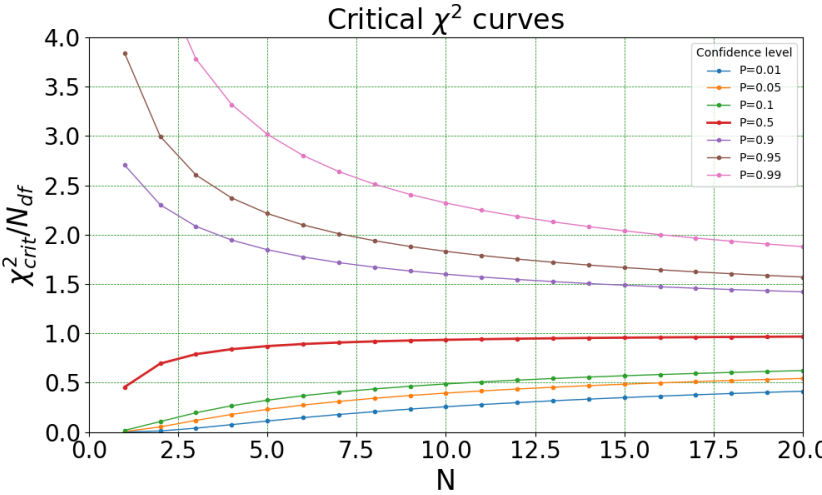
Critical χ^2

07_critical.ipynb Open in Colab

Plot of critical values for reduced χ^2

P=0.5 shows the median

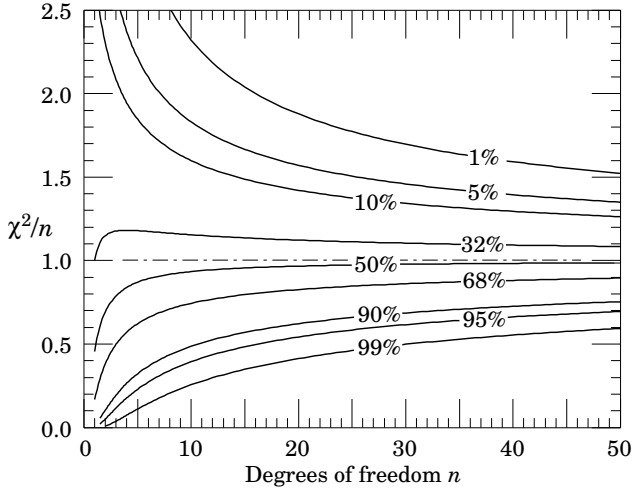
$\chi^2/N_{df} \geq 2$ not excluded for small N



Critical χ^2

Plot of critical values for reduced χ^2

(indicated is $p = 1 - P$)



(PDG)

Student's t Distribution

We can verify consistency of the series of measurements **x** with the true value μ by looking at the shift parameter for the mean

$$z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$$

where mean value $\bar{x}$ is the best estimate of μ assuming Gaussian pdf.

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We need to know measurement uncertainties to calculate χ^2 !...

If the measurement uncertainties are unknown, or not reliable, we can estimate the variance of the sample from the data itself (lecture 04)

$$\hat{s}^2 = \frac{1}{N-1}\sum_i (x_i - \bar{x})^2$$

$\hat{s}^2$ distribution corresponds to χ^2 distribution for $N-1$ degrees of freedom

Student's t Distribution

Consistency of our measurements **x** with the true value μ can be now described by

$$t = \frac{\bar{x} - \mu}{\hat{s}/\sqrt{N}}$$

but the distribution of t is no longer Gaussian (due to $\hat{s}$ being a random variable as well).

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$$f(t; n) = \frac{1}{\sqrt{n\,\pi}} \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}$$

where n is the number of degrees of freedom, $n = N - 1$.

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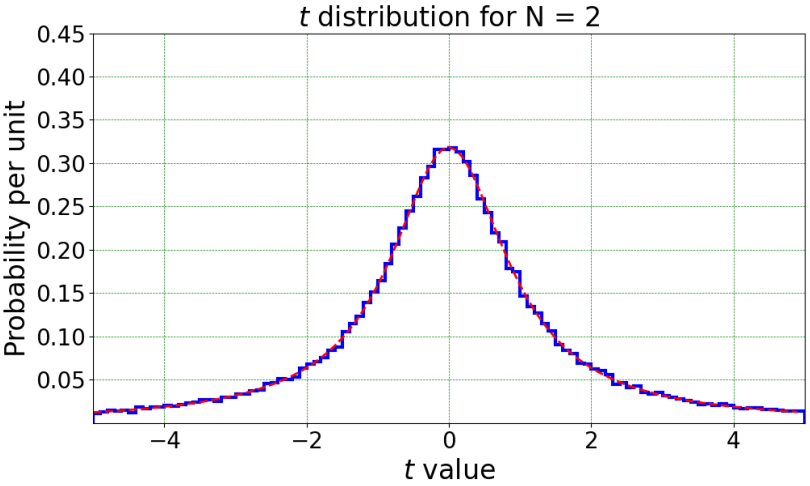
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where n is the number of degrees of freedom, $n = N - 1$.
Distribution is symmetric and has a mean of zero, but larger tails than the Gaussian distribution, for small N in particular.

Student's t Distribution

07_t-dist.ipynb Open in Colab

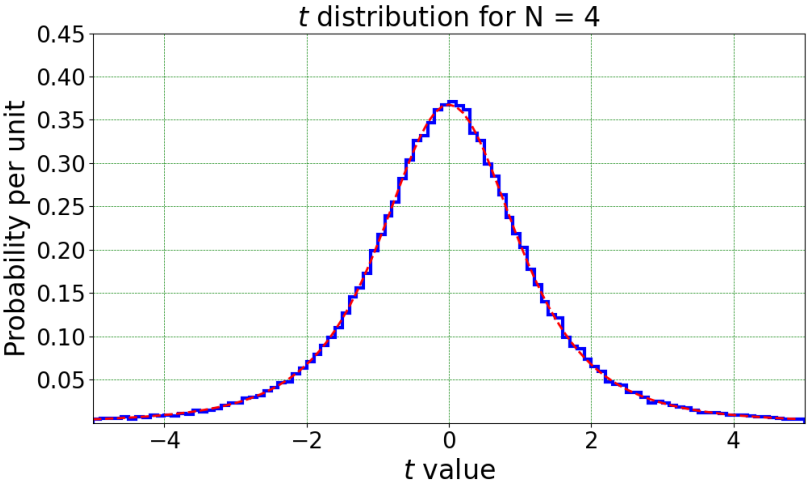


Shape of the t distribution for different numbers of measurements

$N_{df} = N - 1 = 1$

Student's t Distribution

07_t-dist.ipynb Open in Colab

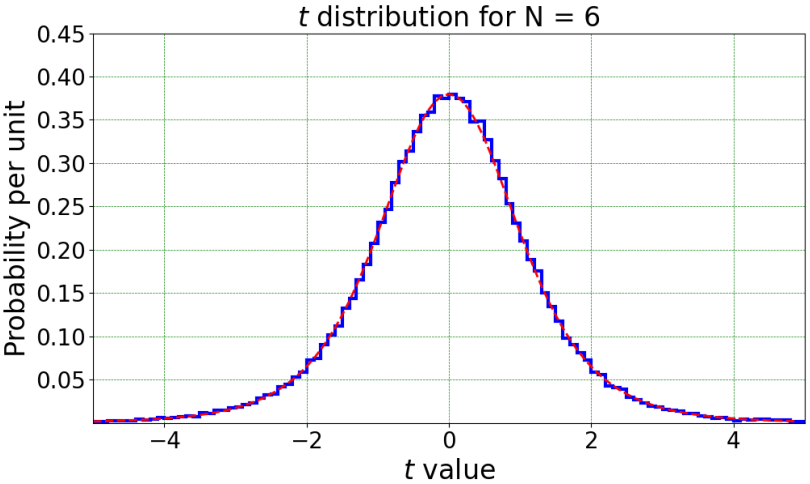


Shape of the t distribution for different numbers of measurements

$N_{df} = N - 1 = 3$

Student's t Distribution

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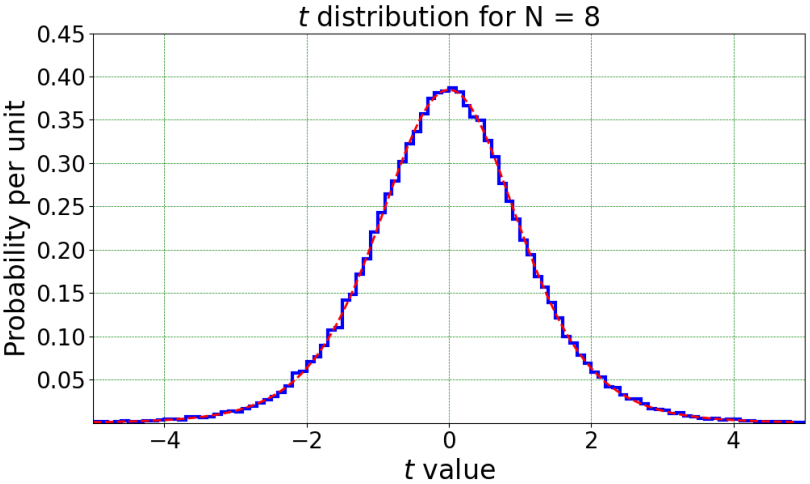


Shape of the t distribution for different numbers of measurements

$N_{df} = N - 1 = 5$

Student's t Distribution

07_t-dist.ipynb Open in Colab

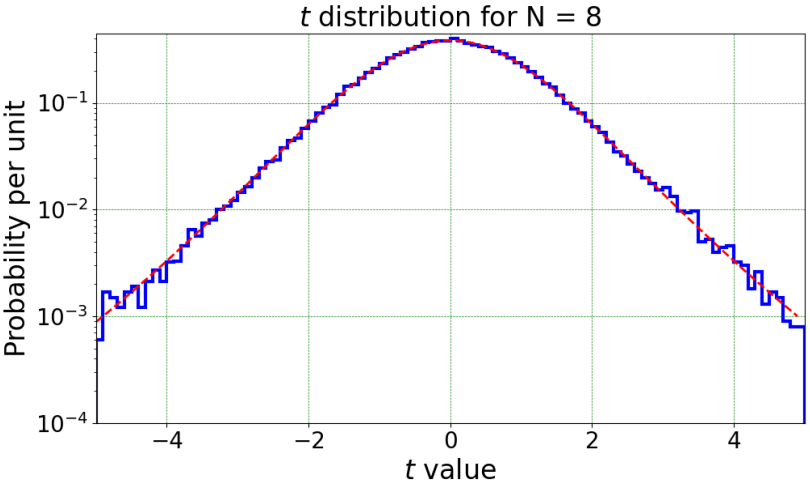


Shape of the t distribution for different numbers of measurements

$N_{df} = N - 1 = 7$

Student's t Distribution

07_t-dist.ipynb Open in Colab



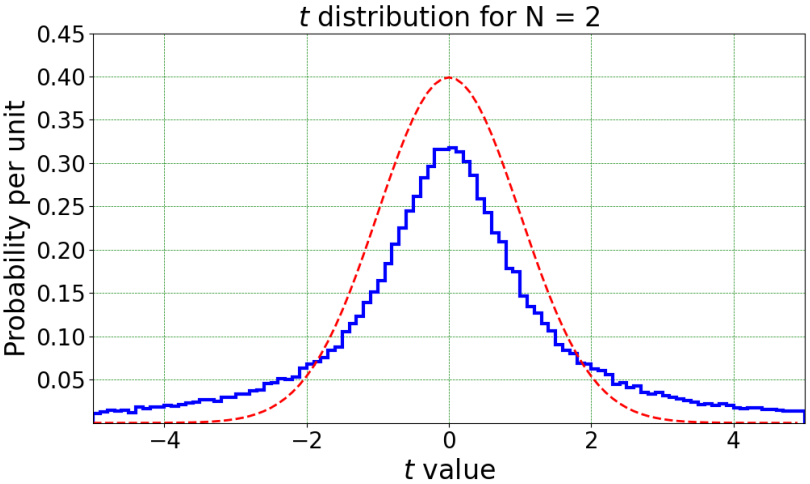
Shape of the t distribution for different numbers of measurements

$N_{df} = N - 1 = 7$

tails are clearly non-Gaussian...

Student's t Distribution

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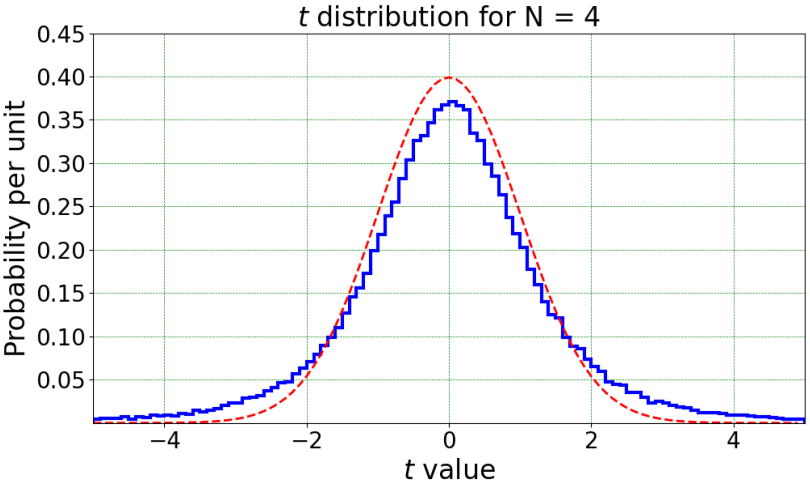


Shape of the t distribution compared with Gaussian distribution

$N_{df} = N - 1 = 1$

Student's t Distribution

07_t-dist.ipynb Open in Colab

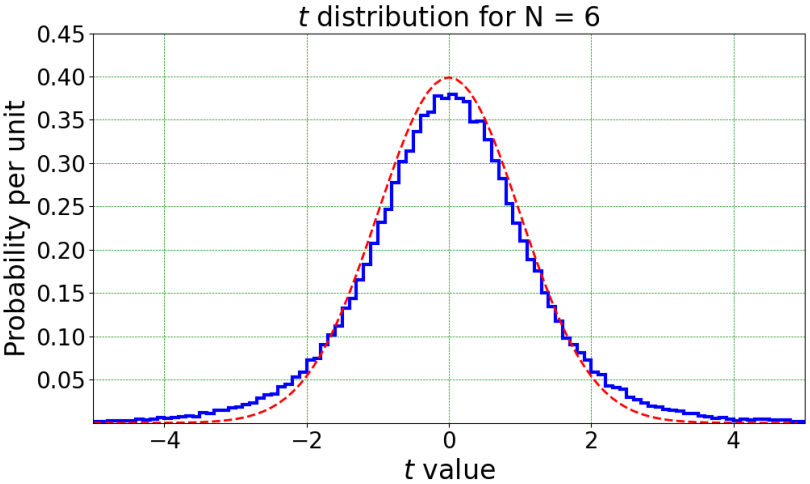


Shape of the t distribution compared with Gaussian distribution

$N_{df} = N - 1 = 3$

Student's t Distribution

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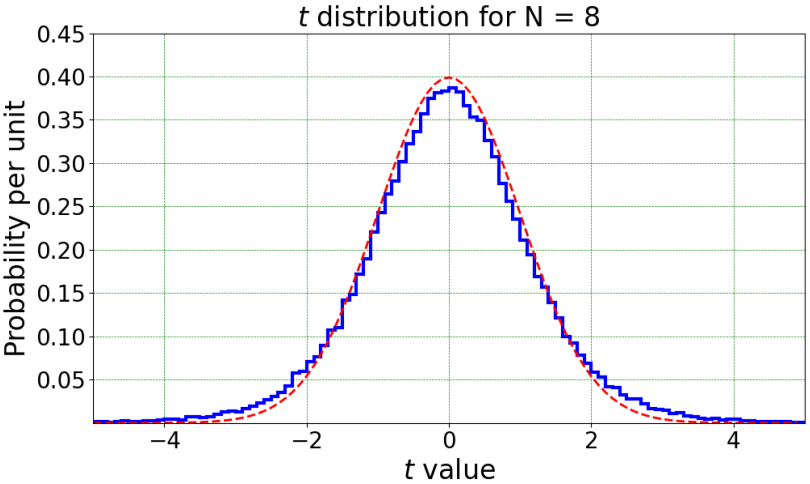


Shape of the t distribution compared with Gaussian distribution

$N_{df} = N - 1 = 5$

Student's t Distribution

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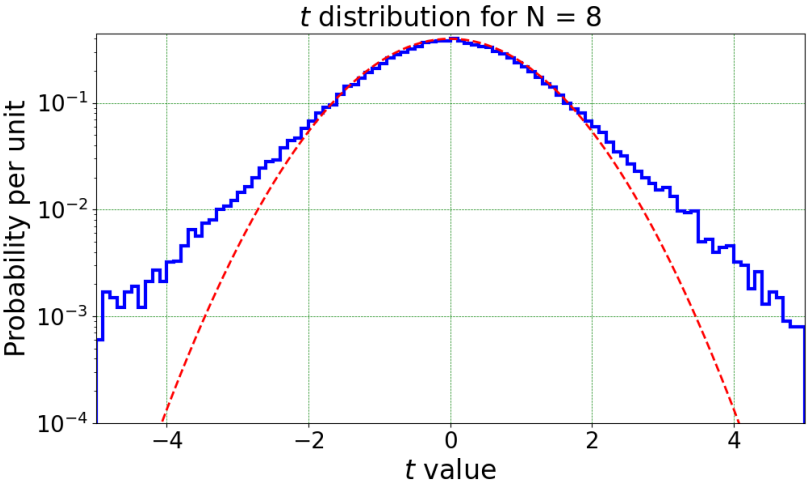


Shape of the t distribution compared with Gaussian distribution

$N_{df} = N - 1 = 7$

Student's t Distribution

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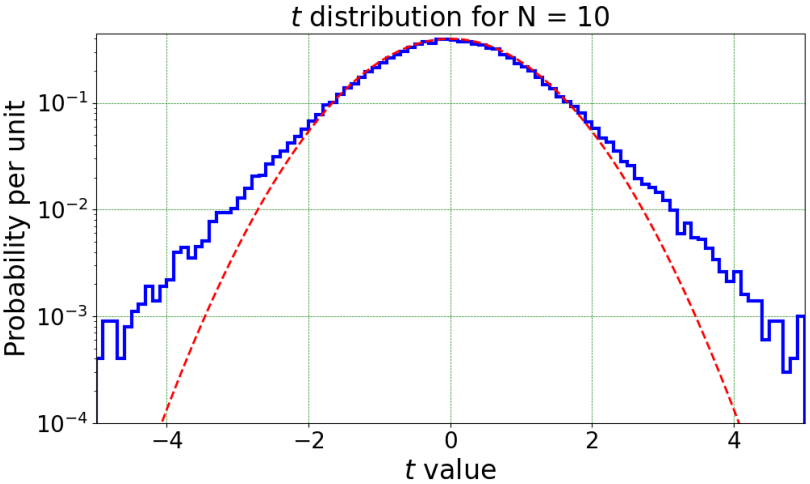


Shape of the t distribution compared with Gaussian distribution

$N_{df} = N - 1 = 7$

Student's t Distribution

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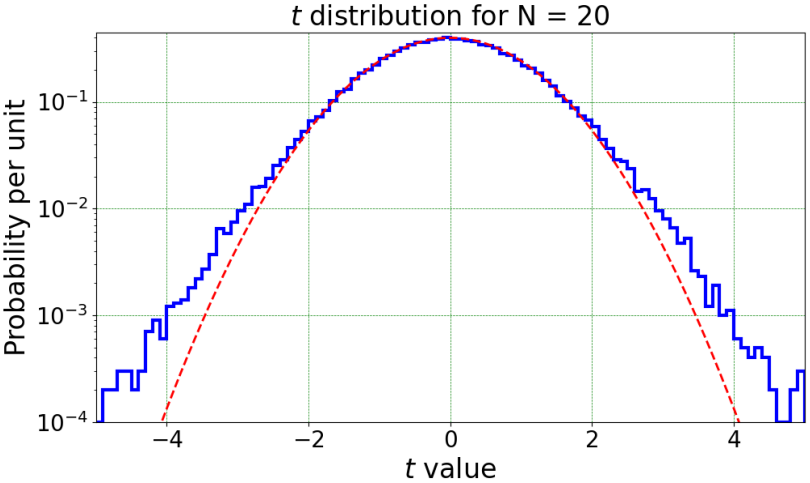


Shape of the t distribution compared with Gaussian distribution

Probability of large fluctuations still significantly enhanced!

Student's t Distribution

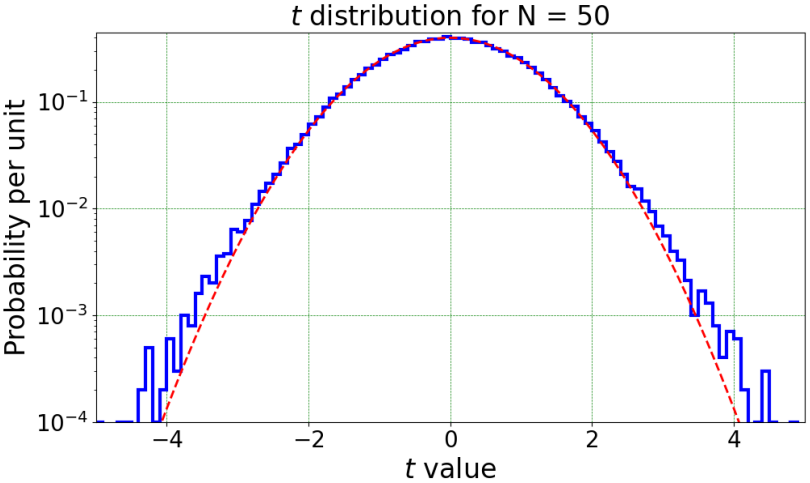
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Shape of the t distribution compared with Gaussian distribution

Student's t Distribution

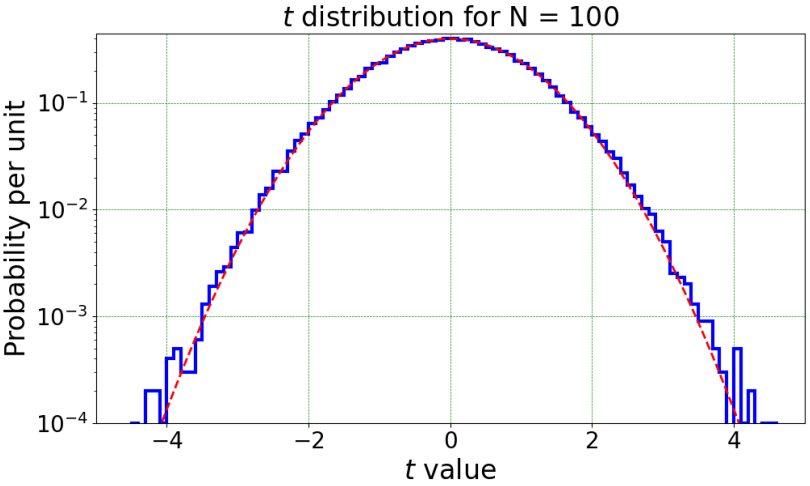
07_t-dist.ipynb Open in Colab



Shape of the t distribution compared with Gaussian distribution

Student's t Distribution

07_t-dist.ipynb Open in Colab



Shape of the t distribution compared with Gaussian distribution

Converges to the Gaussian distribution for large N

Student's t Distribution

(Brandt)

"Critical values" of t for small numbers of degrees of freedom f

$$P = \int_{-\infty}^{t_P} f(t; f)\,\mathrm{d}t$$

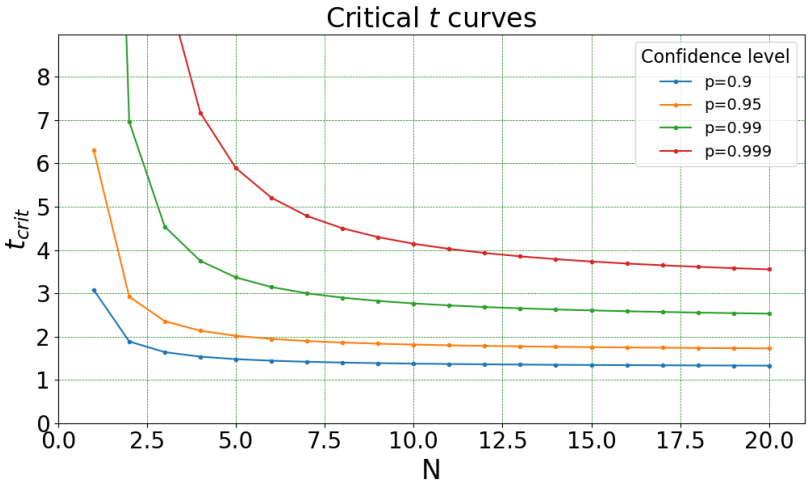
	P						
f	0.9000	0.9500	0.9750	0.9900	0.9950	0.9990	0.9995
1	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140

Student's t Distribution

07_t-limit.ipynb Open in Colab

Plot of critical values for t

Large deviations much more probable for small N



Least-squares method

1. χ^2 distribution
2. Hypothesis Testing
3. Linear Regression
4. Homework

General case

We introduced χ^2 in a very general form:

$$\chi^2 = \sum_{i=1}^{N} \frac{(y_i - \mu_i)^2}{\sigma_i^2}$$

where different μ_i and σ_i are possible for each of N measurement y_i

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$$\chi^2 = \sum_{i=1}^{N} \frac{(y_i - \mu_i)^2}{\sigma_i^2}$$

where different μ_i and σ_i are possible for each of N measurement y_i

It is quite often the case that values of μ_i depend on some controlled variables x_i and a smaller set of model parameters:

$$\mu_i = \mu(x_i; \mathbf{a})$$

we can then use the least-squares method to extract the best estimates of parameters $\mathbf{a}$ from the collected set of data points (x_i, y_i)

We can look for minimum of χ^2 using different numerical algorithms...

Linear case

The case which is particularly interesting is when the dependence is linear in parameters (!):

$$\mu(x;\mathbf{a}) \;=\; \sum_{k=1}^{M} a_k \; f_k(x)$$

where $f_k(x)$ is a set of functions with arbitrary analytical form.

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where $f_k(x)$ is a set of functions with arbitrary analytical form.

One of the examples is the polynomial series:

$$f_k(x) \;=\; x^{k-1} \qquad \Rightarrow \quad \mu(x;\mathbf{a}) \;=\; \sum_{k=1}^{M} a_k \; x^{k-1}$$

but any set of functions can be used, if they are not linearly dependent.
Set of functions ortogonal for a given set of points x_i should work best...

Parameter fit

Bonamente

Least-squares method is the special case of the maximum likelihood approach, when we can assume Gaussian pdf for measurements y_i.

As the best estimate of the parameter set **a** we choose the parameter values which correspond to the (global) χ^2 minimum ($\Rightarrow$ maximum of log-likelihood)

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To look for χ^2 maximum, we consider partial derivatives:

$$\frac{\partial \chi^2}{\partial a_l} = \frac{\partial}{\partial a_l} \sum_{i=1}^{N} \left(\frac{y_i - \sum_{k=1}^{M} a_k \, f_k(x_i)}{\sigma_i} \right)^2 = 0$$

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$$= -2 \sum_{i=1}^{N} \left(\frac{y_i - \sum_{k=1}^{M} a_k \, f_k(x_i)}{\sigma_i^2} \right) f_l(x_i)$$

Parameter fit

Bonamente

We obtain a set of M equations for M parameters a_l: $l = 1 \ldots M$

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or in the matrix form:

$$\mathbb{A} \cdot \mathbf{a} = \mathbf{b}$$

where $\mathbb{A}_{lk} = \sum_{i=1}^{N} \frac{f_l(x_i) \, f_k(x_i)}{\sigma_i^2}$ and $b_l = \sum_{i=1}^{N} \frac{f_l(x_i) \, y_i}{\sigma_i^2}$

Parameter fit

Solution of this set of equations can be obtained by inverting matrix $\mathbb{A}$

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$$\mathbb{C}_{\mathbf{a}} \;=\; \left(-\frac{\partial^2 \ell}{\partial a_l \, \partial a_k} \right)^{-1} = \left(\frac{1}{2} \frac{\partial^2 \chi^2}{\partial a_l \, \partial a_k} \right)^{-1} = \mathbb{A}^{-1}$$

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One can write

$$\mathbb{C}_{\mathbf{a}} \;=\; \left(\sum_{i=1}^{N} \frac{f_l(x_i) \; f_k(x_i)}{\sigma_i^2} \right)^{-1}$$

Expected uncertainties of the extracted parameter values depend on the choice of measurement points x_i but, surprisingly, do not depend on the actual results y_i

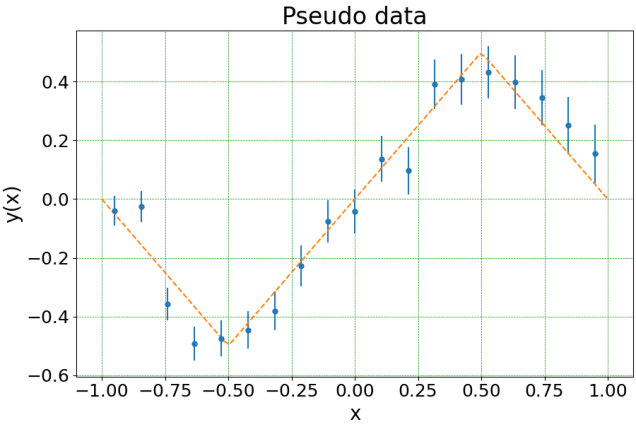
$\Rightarrow$ very useful when planning the experiment...

Linear fit example

07_fit1.ipynb Open in Colab

Fitting Fourier series to example data set

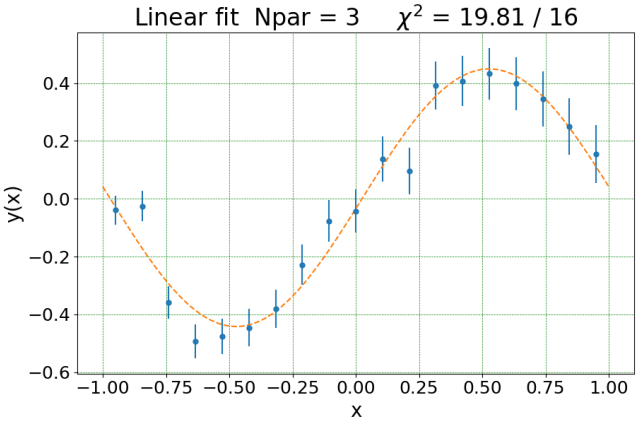
example model ⇒



Linear fit example

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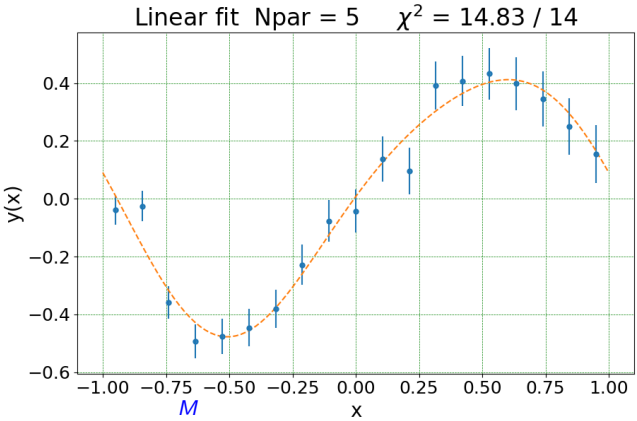


$$y(x) = a_0 + a_1 \sin(x) + a_2 \cos(x)$$

Linear fit example

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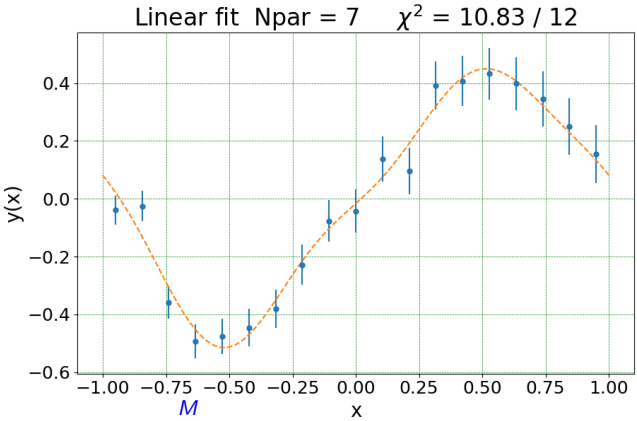
$$y(x) = a_0 + \sum_{n=1}^{M} a_{2n-1}\sin(nx) + a_{2n}\cos(nx) \qquad M = 2$$

Linear fit example

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Fitting Fourier series to example data set

Probably optimal choice



$$y(x) = a_0 + \sum_{n=1}^{M} a_{2n-1}\sin(nx) + a_{2n}\cos(nx) \qquad M = 3$$

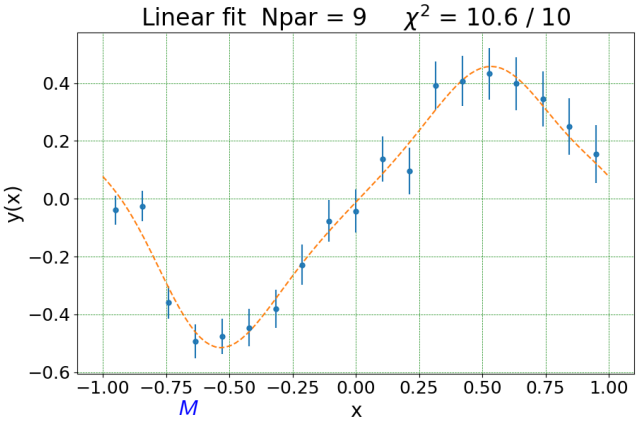
Linear fit example

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Fitting Fourier series to example data set

No improvement...
Alternative choice?



$$y(x) = a_0 + \sum_{n=1}^{M} a_{2n-1}\sin(nx) + a_{2n}\cos(nx) \qquad M = 4$$

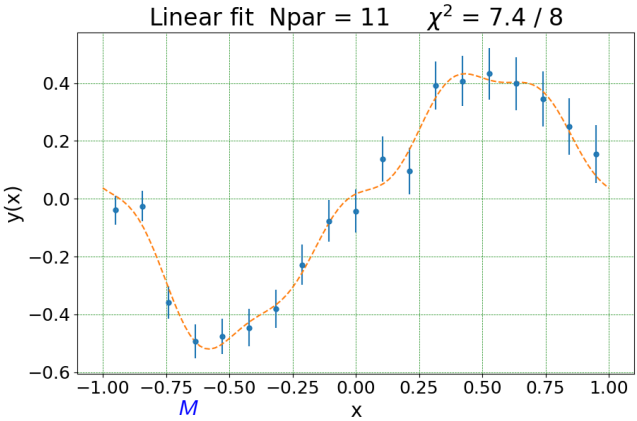
Linear fit example

07_fit1.ipynb

Open in Colab

Fitting Fourier series to example data set

"Overtraining ?"



$$y(x) = a_0 + \sum_{n=1}^{M} a_{2n-1}\sin(nx) + a_{2n}\cos(nx) \qquad M = 5$$

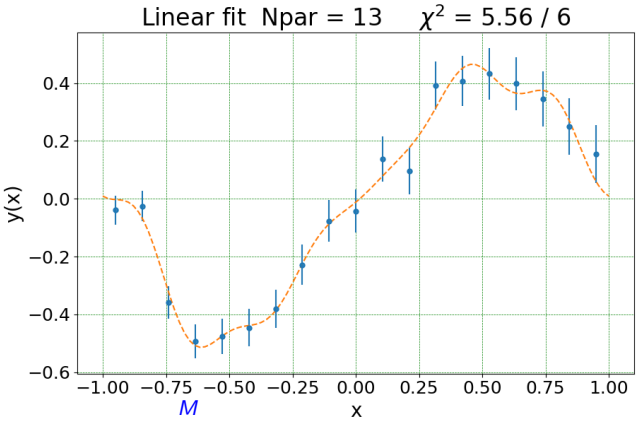
Linear fit example

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Fitting Fourier series to example data set

"Overtraining ?"



$$y(x) = a_0 + \sum_{n=1}^{M} a_{2n-1}\sin(nx) + a_{2n}\cos(nx) \qquad M = 6$$

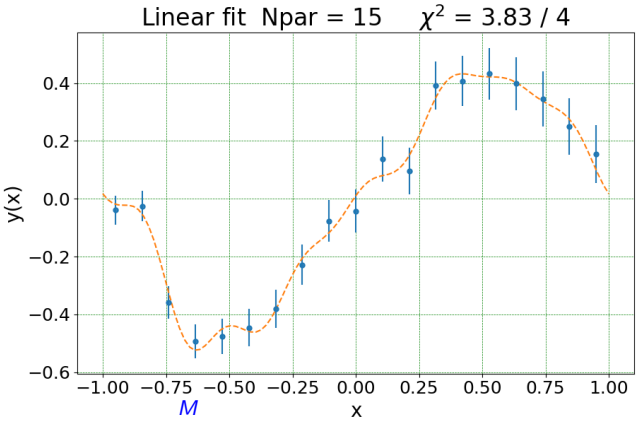
Linear fit example

07_fit1.ipynb

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Fitting Fourier series to example data set

"Overtraining ?"



$$y(x) = a_0 + \sum_{n=1}^{M} a_{2n-1}\sin(nx) + a_{2n}\cos(nx) \qquad M = 7$$

Best fit choice?

If the functional dependence is not predicted by any theory/model, we can try to fit a polynomial or function series. When should we stop?

How can we recognize that we have too many parameters?
Beside looking at the P value corresponding to the obtained χ^2

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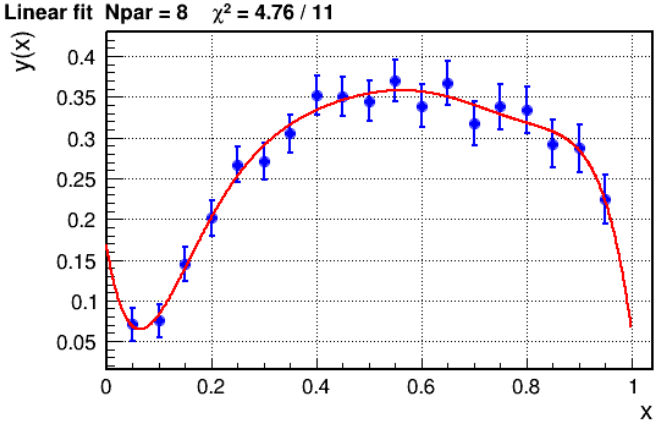
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- Parameters become highly correlated
- Values and errors of the individual parameters increase
  differences of large contributions
- Additional parameters are consistent with zero
- Fit starts to follow fluctuations of the measurement results

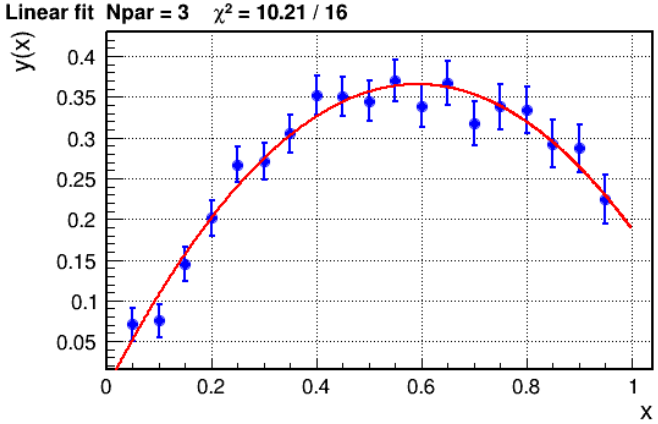
Best fit choice?

Example of fit with too high polynomial order



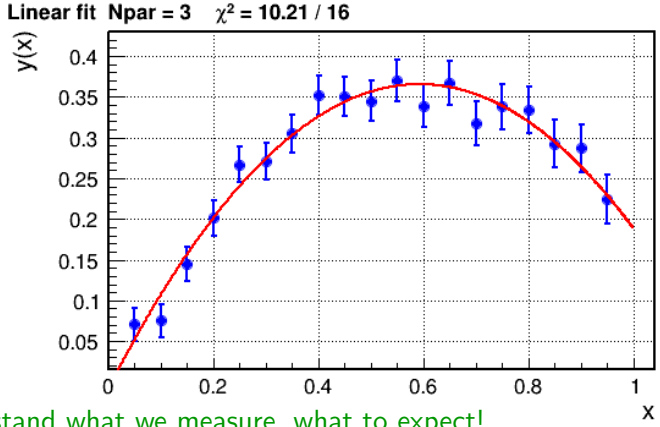
Best fit choice?

Example of fit with proper polynomial order



Best fit choice?

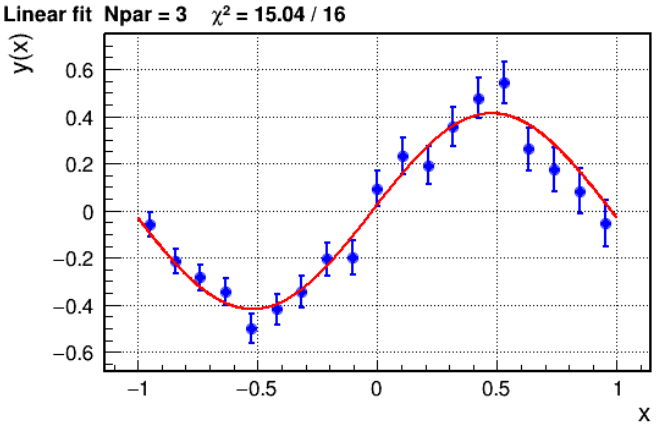
Example of fit with



We need to understand what we measure, what to expect!

Learning on errors

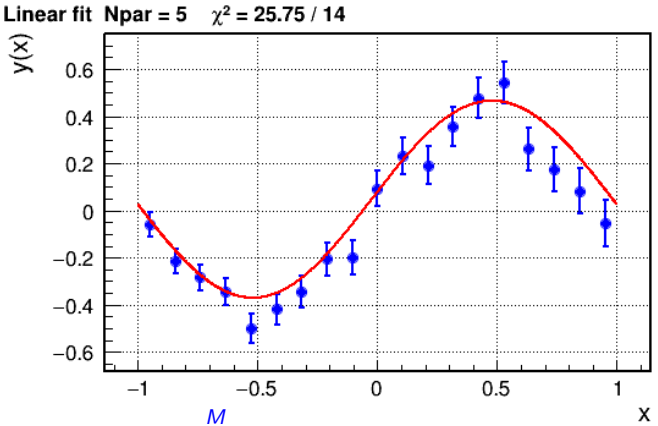
When "wrong" set of functions (highly correlated) is selected...



$$y(x) = a_0 + a_1 \sin(x) + a_2 \cos(x)$$

Learning on errors

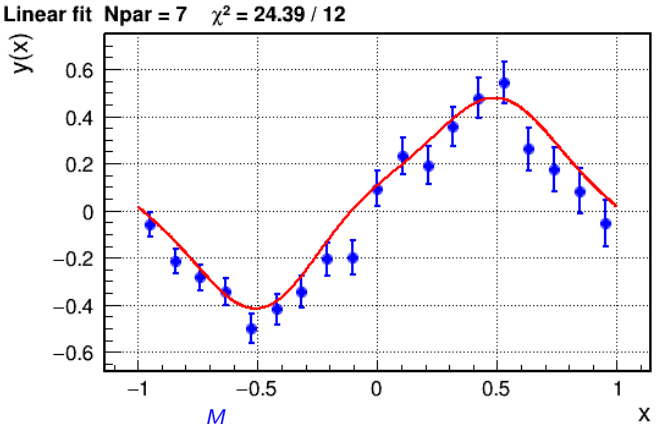
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Learning on errors

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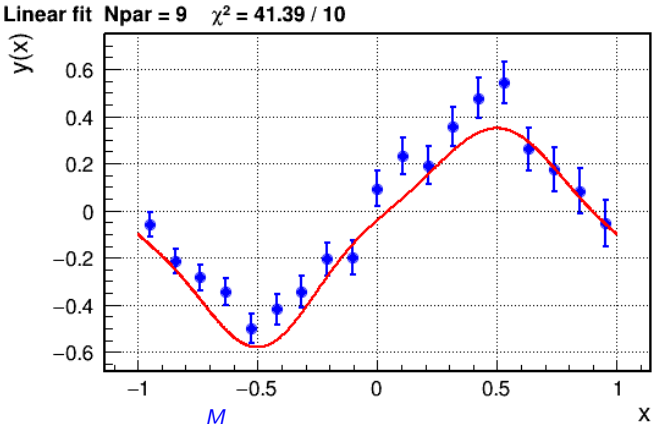


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Learning on errors

When "wrong" set of functions (highly correlated) is selected...

Poor numerical precision due to high correlations between parameters.



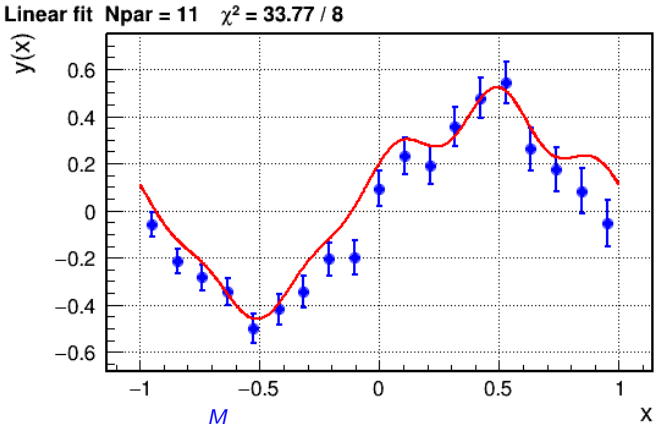
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Learning on errors

When "wrong" set of functions (highly correlated) is selected...

Poor numerical precision due to high correlations between parameters.

$\sin^2(x) + \cos^2(x) = 1$



$$y(x) = a_0 + \sum_{n=1}^{M} a_{2n-1} \sin^n(x) + a_{2n} \cos^n(x) \qquad M = 5$$

Polynomial fit example

For clarity of notation, it is convenient to change parameter numbering to k, $l = 0 \ldots M$ (for polynomial fit of order M, $M+1$ parameters).

$$\mathbb{A}_{l\,k} = \sum_{i=1}^{N} \frac{x_i^{(l+k)}}{\sigma_i^2} \quad \text{and} \quad b_l = \sum_{i=1}^{N} \frac{x_i^l\, y_i}{\sigma_i^2}$$

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For uniform uncertainties it is then:

$$\mathbb{A} = \frac{1}{\sigma^2} \sum_{i=1}^{N} \begin{pmatrix} 1 & x_i & \ldots & x_i^M \\ x_i & x_i^2 & \ldots & x_i^{M+1} \\ \vdots & & & \vdots \\ x_i^M & x_i^{M+1} & \ldots & x_i^{2M} \end{pmatrix} \qquad \mathbf{b} = \frac{1}{\sigma^2} \sum_{i=1}^{N} \begin{pmatrix} y_i \\ y_i\, x_i \\ \vdots \\ y_i\, x_i^M \end{pmatrix}$$

quite simple to implement...

Uncertainty estimate

The χ^2 value at the minimum can be then calculated as:

$$\tilde{\chi}^2 \quad = \quad (\mathbf{y} - \mu(\mathbf{x}; \mathbf{a}))^{\intercal} \, \mathbb{A} \, (\mathbf{y} - \mu(\mathbf{x}; \mathbf{a}))$$

Its distribution should correspond to the χ^2 distribution for $N_{df} = N - M$

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Its distribution should correspond to the χ^2 distribution for $N_{df} = N - M$

If σ is the same for all measurements, the extracted parameter values do not dependent on it!

We can use the calculated value of $\tilde{\chi}^2$ to validate the model (test model hypothesis), but also to "correct" our uncertainties, if we consider them unreliable (or they are unknown).
Resulting variance estimate:

$$\tilde{\sigma}^2 \;=\; \sigma^2 \cdot \frac{\tilde{\chi}^2}{N - M}$$

This is useful in particular when $\tilde{\chi}^2 \ll N_{df}$ (overestimated σ)
For $\tilde{\chi}^2 \gg N_{df}$ we need to consider the possibility that our model is wrong...

Multiple independent variables

The described approach works also for multi-dimensional dependencies!
For example, we can consider polynomial of order M in two coordinates:

$$\mu(x, z; \mathbf{a}) = \sum_{k=0}^{M} \sum_{l=0}^{M} a_{kl}\, x^k\, z^l$$

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All we need to do is to order the pairs of indexes, so that vector $\mathbf{a}$ is properly defined.
Example for $M = 1$ (2-D plane): $\mathbf{a} = (a_{00}, a_{10}, a_{01})$

$$\mathbb{A} = \frac{1}{\sigma^2} \sum_{i=1}^{N} \begin{pmatrix} 1 & x & z \\ x & x^2 & x\,z \\ z & x\,z & z^2 \end{pmatrix} \qquad \mathbf{b} = \frac{1}{\sigma^2} \sum_{i=1}^{N} \begin{pmatrix} y \\ y\,x \\ y\,z \end{pmatrix}$$

where measurement indexes $i = 1 \ldots N$ were skipped for variables x_i, y_i and z_i

Least-squares method

1. χ^2 distribution
2. Hypothesis Testing
3. Linear Regression
4. Homework

Homework

Solutions to be uploaded by December 2.

Download the set of data from the lecture home page.
Text file with three columns: x_i, y_i, σ_{y_i}

Use linear regression method to fit polynomial dependence to the data.

Calculate p-value for the 3rd and 4th order polynomial fit.

Point to the order of polynomial, which is adequate for the description of the data and give arguments for your choice.

